Chains, Clearings, and Rebels: Solving a Family of Mode-Switching Subtraction Games

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Abstract

Mode-switching subtraction games are non-invariant subtraction games whose legal moves depend on the pile size n modulo a fixed m. The diagonal family $D(m,r)$ has two ingredients: from positions $n \equiv r \pmod{m}$ a player subtracts any positive square at most n, and from every other position exactly m. For every $m \geq 3$ except $m = 4$ and every r, the P-positions of $D(m,r)$ are precisely the $m-1$ residue classes modulo $2m$ of the chain bottoms, together with $n = 0$ and, for $(m,r) \in \{(9,4),(9,5),(9,6)\}$, a single rebel position $n = m + r \in \{13,14,15\}$; the law holds for both parities of m with no exceptions beyond these named positions. At $m = 2$ and $m = 4$ the squares modulo $2m$ lie inside $\{0,1,m\}$, so no move from the affected classes reaches a foreign P-position: those classes are confined to self-contained recursions, which are Golomb's subtract-a-square game in affine disguise: confinement is proved, and the classes are provably never eventually periodic, of density zero within their class. The same architecture solves Foursquare, an earlier machine-found game with moves $\{3,8,9\}$ and a square mode at $n \equiv 1 \pmod{4}$: its P-positions are $n \equiv 0,2,4,6 \pmod{16}$, and enlarging the move set to the full residue classes of 3, 8, and 9 modulo 16 provably leaves the P-set unchanged. Every conjecture was machine-proposed from a census of roughly 3,000 games; every proof is human-written; twice, theory constructed afterwards rederived numbers the instrument had recorded before the theory existed. The growth rate of their P-sets — Golomb's own open question since 1966 — remains open.

1 Introduction

A subtraction game is played on a pile of counters: a ruleset names the amounts a player may remove, the players alternate, and under normal play whoever cannot move loses. For a fixed finite ruleset the classical theory ends in one word, periodicity: the sequence of previous-player wins is eventually periodic, and the game solves itself by a finite computation [2]; see [3] for a recent survey. The live frontier is rulesets in which the set of legal moves varies with the position being moved from — position-dependent, or non-invariant, move sets [1]. This paper studies a structured family at that frontier, in which the entire mode of play — which amounts are legal, and why — switches with the residue of n.

The protagonist is the two-parameter family $D(m,r)$: from a position $n \equiv r \pmod{m}$ a player may subtract any positive square not exceeding n; from every other position the only move is to subtract m. The family is called diagonal because its members first surfaced along a diagonal of the census's parameter grid, one game per modulus. The main result can be tasted before any machinery arrives: for every $m \geq 3$ except $m = 4$ and every r, the P-positions of

$D(m, r)$ are exactly $m-1$ residue classes modulo $2m$ — the bottoms of the game's fallback chains — together with a completely determined finite exception set, for both parities of m and with no unverified thresholds.

The contributions come in the paper's own order. The Chain Lemma decomposes every fallback class into an independent chain whose P-positions alternate upward from a stuck bottom, reading the law's residues directly off those bottoms. The Clearing Lemma turns one small square per class into a complete escape argument, and with it comes the Diagonal Theorem: every $m \geq 3$ except $m = 4$, both parities, the exception set pinned to exactly three rebel positions and one boundary. Confinement explains the excluded moduli: at $m = 2$ and $m = 4$ the squares modulo $2m$ collapse into $\{0, 1, m\}$, no move reaches a foreign P-position, and the affected classes turn inward into self-contained recursions — the census's two lawless columns, identified. Section 7 carries the architecture to Foursquare, the machine's first find: its law $n \equiv 0, 2, 4, 6 \pmod{16}$ proved in half a page, its mechanism of one governor and paired rescuers exposed, and Kadam's Extension established. Section 8 opens the instrument itself — proposer, engine, detectors, filter — and records the two retrodictions in which theory built afterwards rederived numbers recorded before the theory existed.

The method deserves a plain statement on page one. Every conjecture in this paper was machine-proposed, drawn from a census of roughly three thousand games; every proof is human-written. Section 8 documents the pipeline, the verification regime, and the written integrity protocol that kept the two roles separate.

Section 2 fixes notation and the verification lemma; Sections 3 through 5 build chains, clearings, and rebels into the Diagonal Theorem; Section 6 proves confinement at $m = 2$ and 4; Section 7 puts the machinery to work three times — $D(5, 2)$ solved in full as the worked example, Foursquare with its mechanism and Kadam's Extension, and the odd-fallback collapse; Section 8 opens the instrument; and Section 9 closes the question the census could not settle — and locates the one that remains, older than this paper.

2 Preliminaries

Definition 2.1 (Positions and outcomes)**.** A terminal position (no legal moves) is a P-position; a position is an N-position if and only if at least one legal move reaches a P-position; a position is a P-position if and only if every legal move reaches an N-position, the terminal case satisfying this clause vacuously.

Lemma 2.2 (Verification principle; see, e.g., [9])**.** *Let S be a set of positions of an impartial game under normal play. If (i) every terminal position lies in S, (ii) no move from a position in S leads to a position in S, and (iii) from every position not in S some move leads to a position in S, then S is exactly the set of* P*-positions.*

The general theory of finite subtraction games and their eventual periodicity is developed in [4].

Definition 2.3 (The diagonal games $D(m, r)$)**.** Let $m \geq 2$ and $0 \leq r \leq m - 1$. $D(m, r)$ is played on the non-negative integers. From a position $n \equiv r \pmod{m}$ (*squares mode*) a move subtracts any square k^2 with $k \geq 1$ and $k^2 \leq n$. From any other position (*fallback mode*) the only move subtracts m, available when $n \geq m$. Play is normal.

Definition 2.4 (Squares modulo $2m$)**.** $\mathcal{Q}(2m)$ denotes the set of residues of squares modulo $2m$. The mirror tool, proved once in general: $(2m-k)^2 = 4m^2 - 4mk + k^2 \equiv k^2 \pmod{2m}$, since $4m^2 = 2m \cdot 2m$ and $4mk = 2k \cdot 2m$. So every table is $k = 1..m$ plus mirrors, plus the 0 from $k = 2m$. Two side facts fell out of the mirrors and are used below: for odd m, $m^2 - m = m(m-1)$ has the even factor $m-1$, so $m^2 \equiv m \pmod{2m}$; for even m, $m^2 = (m/2) \cdot 2m \equiv 0 \pmod{2m}$.

mod	$\mathcal{Q}(2m)$	complement
4	$\{0,1\}$	$\{2,3\}$
6	$\{0,1,3,4\}$	$\{2,5\}$
8	$\{0,1,4\}$	$\{2,3,5,6,7\}$
10	$\{0,1,4,5,6,9\}$	$\{2,3,7,8\}$
12	$\{0,1,4,9\}$	$\{2,3,5,6,7,8,10,11\}$
14	$\{0,1,2,4,7,8,9,11\}$	$\{3,5,6,10,12,13\}$
18	$\{0,1,4,7,9,10,13,16\}$	$\{2,3,5,6,8,11,12,14,15,17\}$
22	$\{0,1,3,4,5,9,11,12,14,15,16,20\}$	$\{2,6,7,8,10,13,17,18,19,21\}$

Table 1: The engines at every modulus: $\mathcal{Q}(2m)$ with complements.

The mod-4 and mod-8 rows are the indictment written out: $\mathcal{Q}(4) = \{0,1\}$ and $\mathcal{Q}(8) = \{0,1,4\}$ both sit entirely inside the triple $\{0,1,m\}$, so their complements swallow every value any hard set will ever contain (Section 4) — the disease of $m=2$ and $m=4$ is visible in these two rows before any game logic starts. One eyebrow kept raised on purpose: the thinnest *odd* table is mod 18, thinned because 9 is itself a square and the rows $k=3$ and $k=9$ collide there — $m=9$'s anomaly is foreshadowed in its own table.

3 The Chain Lemma

Lemma 3.1 (Chain Lemma)**.** *Fix integers $m \geq 2$ and $0 \leq r \leq m-1$, and let c be any residue with $0 \leq c \leq m-1$ and $c \neq r$. In $D(m,r)$, for every position $n \geq 0$ with $n \equiv c \pmod{m}$:*

$$n \text{ is a P-position} \iff n \equiv c \pmod{2m}.$$

Proof. Rungs. The positions of class c form the progression c, $c+m$, $c+2m$, $\ldots$; call $c+jm$ rung j. *Closure.* A rung is never $\equiv r \pmod{m}$, so its only allowed move is the fallback $-m$, legal exactly when $j \geq 1$ and landing on rung $j-1$; rung $0 = c < m$ has no move at all, and no move ever leaves the progression. *Independence.* The status of a position is defined recursively from its options alone — the definition never mentions which positions move in. Since Closure makes every option of a rung another rung, the chain computes its statuses as if the rest of the game had been deleted. *Alternation.* We claim rung j is P if and only if j is even. Base: rung 0 is terminal by Closure, hence P. Step: assume the claim for rung $j-1$; rung j's single option is rung $j-1$, and a single-option position is P if and only if its option is N — so for j even, rung $j-1$ is N by the assumption and rung j is P, while for j odd, rung $j-1$ is P and rung j is N. *Translation.* j is even exactly when $c + jm \equiv c \pmod{2m}$, so the P-positions of class c are precisely the $n \equiv c \pmod{2m}$. ∎

4 Escape: the Clearing Lemma and its exceptions

Lemma 3.1 classifies all positions outside the square-mode class $r \pmod m$, but it is silent about positions satisfying $n \equiv r \pmod m$; the purpose of this section is to determine their outcomes. Since this class splits into the two residue classes r and $r+m \pmod{2m}$, we treat them separately. The first, which we call the free residue, has an immediate move to a P-position from Lemma 3.1, while the second, the hard residue, requires a suitable square move to reach such a P-position. The Clearing Lemma identifies when such a move exists, and the exceptional cases arise precisely when this clearing mechanism fails.

Definition 4.1 (Window; free and hard residues; hard set)**.** By Lemma 3.1, each class $c \neq r$ is a self-contained chain whose P-positions are exactly $n \equiv c \pmod{2m}$. The window $T(m,r)$ is the set of foreign bottoms mod $2m$. Class r splits mod $2m$ into the **free residue** (r when $r \geq 1$, else m) and the **hard residue** ($r+m$ when $r \geq 1$, else 0). The free escape is immediate: 1^2 lands the free residue on the bottom directly beneath it ($r-1$, or $m-1$ when $r=0$), and is legal from the smallest free member. For the hard residue, a square value $s \pmod{2m}$ lands in the window iff

$$s \in H(m,r) = \{r+1,\ldots,m-1\} \cup \{m+1,\ldots,m+r\} \text{ for } r \geq 1,$$
$$\text{or} \quad \{m+1,\ldots,2m-1\} \text{ for } r=0.$$

Remark 4.2 (Constitutional uselessness). For every m and r: $0, 1, m \notin H(m,r)$; a square $\equiv 0$ or $m \pmod{2m}$ is $\equiv 0 \pmod m$ and keeps a class-r position inside class r; and subtracting a square $\equiv 1 \pmod{2m}$ from the hard residue lands one below it in the top half, an N-position.

Proof. The exclusions $0, 1, m \notin H(m,r)$ are read directly off the displayed form of H. A square $\equiv 0$ or $m \pmod{2m}$ is $\equiv 0 \pmod m$, so subtracting it keeps a class-r position inside class r; and subtracting a square $\equiv 1 \pmod{2m}$ from the hard residue lands one below it in the top half — the odd rung directly above a bottom — an N-position. □

Lemma 4.3 (Clearing Lemma)**.** *Suppose some square $s \neq m$ lies in the integer interval $(r,\, m+r]$ when $r \geq 1$, or in $(m,\, 2m)$ when $r=0$. Then the entire hard class is* N *by the single move $-s$.*

Proof. Legality: $s \leq m+r$, the smallest hard member (for $r=0$: $s < 2m$, and the smallest positive hard member is $2m$), so the move is affordable to every member from the first. Landing: $n - s \equiv m + r - s \pmod{2m}$ is a foreign bottom — in (r,m) when $r < s < m$, in $[0,r)$ when $m < s \leq m+r$; for $r=0$, the landing $2m-s$ lies in $(0,m)$. Combined with the free escape, class r is all N — except possibly $n=0$, handled in Lemma 4.7. □

Lemma 4.4 (The interval contains a square)**.** *For $m \geq 3$ and $1 \leq r \leq m-1$, the interval $(r,\, m+r]$ contains a square. For $r=0$, the interval $(m,\, 2m)$ contains a square for every $m \geq 3$ except $m=4$.*

Proof. For $r \geq 1$: if $(r, m+r]$ contained no square, consecutive squares would straddle it, $c^2 \leq r$ and $(c+1)^2 > m+r$, so $2c+1 > m$, hence $c \geq m/2$ for any m, hence $c^2 \geq m^2/4 > m-1 \geq r$, since $m^2/4 - (m-1) = (m-2)^2/4 > 0$ for $m \geq 3$ — contradicting $c^2 \leq r$. The inequality degenerates at exactly one modulus, $m=2$, where $(m-2)^2 = 0$ and the interval can indeed be empty ($(1,3]$ in $D(2,1)$ contains no square). Suppose the interval $(m, 2m)$ contained no square. Let c be the largest integer with $c^2 \leq m$; then $(c+1)^2 \geq 2m$, and subtracting gives $2c+1 \geq 2m-m = m$, so $c \geq (m-1)/2$. Squaring against $c^2 \leq m$ yields $(m-1)^2/4 \leq m$, i.e. $m^2 - 6m + 1 \le 0$, so $m < 6$. The remaining cases are checked by hand: $(3, 6)$ contains 4 and $(5, 10)$ contains 9, while $(2, 4)$ and $(4, 8)$ contain no square. Hence the interval is nonempty for every $m \ge 3$ except $m = 4$, and the two exceptions are exactly the degenerate moduli of Section 6. ☐

Proposition 4.5 (The failure window)**.** *The hypothesis of Lemma 4.3 fails for $r \ge 1$ only if $m = b^2$ with $(b-1)^2 \le r \le 2b$, which is possible only for $b \le 3$: among $m \ge 3$, exactly $m = 4$ ($r \in \{1, 2, 3\}$) and $m = 9$ ($r \in \{4, 5, 6\}$).*

Proof. With a square always present, the Clearing Lemma's hypothesis can fail for $r \ge 1$ only if the *unique* square in $(r, m+r]$ is m itself: $m = b^2$ with $(b-1)^2 \le r$ (no smaller square intrudes) and $(b+1)^2 > m + r$, i.e. $r \le 2b$. The window $(b-1)^2 \le r \le 2b$ is nonempty iff $(b-1)^2 \le 2b$ iff $b \le 3$. Among $m \ge 3$ that leaves exactly $m = 4$ ($b = 2$, $r \in \{1, 2, 3\}$) and $m = 9$ ($b = 3$, $r \in \{4, 5, 6\}$). With the $r = 0$ erratum, every class of $m = 4$ is condemned from the outside — no clearing square exists for any r — while $m = 9$ is healthy at every r except $4, 5, 6$. Squarehood of m alone is harmless: at $m = 25$ ($b = 5$) the neighboring square 36 sits only $2b+1 = 11$ above m, deep inside a 25-wide window, so it invades $(r, m+r]$ for every r; a square m isolates itself only when the modulus is small enough ($b \le 3$) for square-gaps to outrun m. ☐

Proposition 4.6 (The rebels (exceptional positions))**.** *For $(m, r) \in \{(9, 4), (9, 5), (9, 6)\}$ the position $n = m + r$ is a* P*-position, and it is the unique* P*-position of $D(m, r)$ outside the window.*

Proof. At $(9, r)$ with $r \in \{4, 5, 6\}$, the smallest hard member $n_0 = m + r \in \{13, 14, 15\}$ can afford only the squares $k^2 \le r$ and 9. A square $k^2 \le r$ lands on $m + (r - k^2)$, the odd rung directly above bottom $r - k^2$ — N. The square 9 lands on r itself, the smallest free member — N, since it escapes by 1^2. Every move lands on N, so n_0 **is** P**.** In-game receipts: in $D(9, 4)$, $13 \to 12, 9, 4$, all N; in $D(9, 5)$, $14 \to 13, 10, 5$, all N; in $D(9, 6)$, $15 \to 14, 11, 6$, all N. Above the rebel the class closes uniformly: the members $27+r$ $(31, 32, 33)$ are cleared by $25 \equiv 7 \pmod{18}$, which lies in all three hard sets, landing on bottoms $6, 7, 8$; the members $45 + r$ $(49, 50, 51)$ by $49 \equiv 13$, landing on bottoms $0, 1, 2$; and from $63 + r \ge 64 = (m-1)^2$ onward, the witness $64 \equiv 10 = m + 1$ lands on bottom $r - 1$ forever. So each exceptional game carries exactly one extra P-position, and the exception set is precisely the set E of Theorem 5.1. ☐

Lemma 4.7 (Boundary positions corrupt nothing)**.** *In $D(m, r)$: (i) if $r = 0$ then $n = 0$ is a* P*-position, and its status alters no other position's; (ii) for $(m, r) \in E$, the* P*-position $n = m+r$ alters no other position's status.*

Proof. **Corruption audits.** The $n = 0$ clause: when $r = 0$, zero is claimed by squares mode and is the unique squares-mode position that can afford no square ($k \ge 1$ forces $k^2 \ge 1 > 0$), so it is terminal, hence P — while the unamended law would file residue 0 as N. It corrupts nobody: fallback chains never visit class 0; the positions that can drop to 0 are $n = k^2 \equiv 0 \pmod{m}$, class-0 members the theorem already declares N; and the classes above escape without it — spot receipts: $10 - 9 = 1$ (P) in $D(5, 0)$, $14 - 9 = 5$ (P) in $D(7, 0)$. For $r \ge 1$ the clause is absorbed, since 0 is itself a foreign bottom; it earns its keep exactly at $r = 0$. The rebels, one sentence in the same mirror: a rebel corrupts nobody either — fallback chains never enter class r, and the only positions with a move onto $m + r$ are class-r members above it (a square step back into class r forces $k^2 \equiv 0 \bmod 9$, i.e. $3 \mid k$), which the theorem already declares N, so the rebel's P-ness merely hands them one more witness for what they already were. ☐

5 The Diagonal Theorem

The pieces are now on the table — chains classifying every class except r, clearings settling class r wherever a suitable square exists, the failures catalogued down to three rebels and one boundary — and Theorem 5.1 assembles them into the classification.

Theorem 5.1 (Diagonal Law)**.** *Let $m \geq 3$ with $m \neq 4$, and let $0 \leq r \leq m-1$. Define $E = \{(9,4),(9,5),(9,6)\}$. In $D(m,r)$, a position n is a* P*-position if and only if*

$$n \equiv b \pmod{2m} \text{ for some foreign bottom } b \in \{0,1,\ldots,m-1\} \setminus \{r\},$$
$$\text{or} \quad n = 0, \quad \text{or} \quad (m,r) \in E \text{ and } n = m+r.$$

The law holds for both parities of m, with no thresholds and no unverified zones.

Proof. Fix a position n of $D(m,r)$ with $m \geq 3$, $m \neq 4$: either n lies in a class $c \not\equiv r \pmod{m}$, or n lies in class r, and in the latter case n is either a boundary position named in the statement — $n = 0$ when $r = 0$, or $n = m + r$ when $(m,r) \in E$ — or an ordinary class-r position, sitting by Definition 4.1 on either the free or the hard residue modulo $2m$; these buckets exhaust all n, and we take them in order. *Case 1 (foreign classes).* If n lies in a class $c \not\equiv r$, then by Lemma 3.1 (the Chain Lemma) its chain is self-contained and n is P exactly when $n \equiv c \pmod{2m}$ — precisely the foreign-bottom clause of the statement, so every such position carries its verdict. *Case 2 (free residue).* If n sits on the free residue, the move -1^2 is legal from the smallest free member onward and lands on the foreign bottom directly beneath ($r-1$, or $m-1$ when $r=0$), a P-position *by the chain verdict of the previous sentence* (Lemma 3.1), so every free-residue position is N — and indeed the statement names no free residue among its P-positions. *Case 3 (hard residue, escaped).* If n sits on the hard residue and $(m,r) \notin E$, then Lemma 4.4 together with Proposition 4.5 supplies a square $s \neq m$ in the clearing interval (the hypothesis $m \neq 4$ is load-bearing precisely here, since at $m = 4$ and only there they offer nothing for any r), and Lemma 4.3 converts that one square into the verdict: every hard-residue position is N by the single move $-s$, for both parities of m, the boundary $n = 0$ having already been set aside. *Case 4 (the exceptional games).* If n is a class-r position with $(m,r) \in E$, Proposition 4.6 delivers the exceptional verdict: $n = m + r$ is P and is the unique off-window P-position, every other hard member being N, which combined with the free verdict above is exactly the exception clause of the statement. *Case 5 (boundaries).* Finally, Lemma 4.7 certifies the boundary clauses and their harmlessness: $n = 0$ at $r = 0$ is terminal hence P, and neither its P-ness nor a rebel's disturbs any verdict already issued — every position with a move onto them is a class-r position ruled N above, and no fallback chain ever enters class r — so the classifications of the four preceding sentences stand undisturbed, and the P-positions of $D(m,r)$ are exactly those of the statement: the foreign bottoms modulo $2m$, together with $n = 0$, together with $n = m+r$ when $(m,r) \in E$. ☐

Remark 5.2 (The historical witness). *The odd-m witness identity, the original route.* For odd m, $(m-1)^2 = m^2 - 2m + 1 \equiv m^2 + 1 \equiv \mathbf{m+1} \pmod{2m}$, using $m^2 \equiv m$ from Definition 2.4; and $m+1$ belongs to every hard set, landing the hard residue exactly on bottom $r-1$ ($m-1$ when $r=0$). So the pair $(1^2, (m-1)^2)$ is a universal witness pair for all odd m at once — at the price of the threshold $n \geq (m-1)^2$. The Clearing Lemma supersedes it: a square below $m+r$ does the same job with legality free, for both parities, with no threshold. The identity remains the cleanest line in this development and the historical door to everything above.

6 The diseased (degenerate) moduli

Theorem 5.1 excludes exactly two moduli, and this section proves the exclusion is not a limitation of the method but a fact about squares: at $m = 2$ and $m = 4$ there is no rescue to find, and the affected classes turn inward.

Lemma 6.1 (Confinement Lemma)**.** *For $m \in \{2, 4\}$ and every r: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$, while $H(m, r)$ excludes 0, 1, and m (Remark 4.2). Hence no square move from a hard-residue position reaches the window: every move stays inside class r or lands on an* N*-position of a foreign chain.*

Proof. For $m \in \{2, 4\}$ and every r: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$ — read directly off the mod-4 and mod-8 rows of Table 1 — while every hard set $H(m, r)$ excludes 0, 1, and m (Definition 4.1). Hence from any hard-residue position, every legal square move either stays inside class r (a square $\equiv 0$ or m mod $2m$ is $\equiv 0$ mod m) or lands one below on the odd rung above a bottom (a square $\equiv 1$), an N-position — never on a P-position of a foreign chain. The argument is an only-options enumeration over $\mathcal{Q}(2m)$, the mirror image of the Independence step in the proof of Lemma 3.1 — there, fallback moves cannot leave a chain; here, square moves cannot leave the class except onto N. *Consequence:* the hard class's P/N structure is decided by a self-contained internal recursion, and its very first member already defies the chain law: 3 is P in $D(2, 1)$ and 5 is P in $D(4, 1)$, where escape, if it existed, would force N. Calibration: for $m = 2$ and $m = 4$ we prove confinement — no square door into the window, so the class turns inward. We do **not** prove that these confined classes obey no eventual law of their own: "no period and no modulus up to 200 fits" is a reading from the search instrument, not a theorem, and their aperiodicity remains open. □

Remark 6.2 (Double condemnation)*.* The two diseased moduli are exactly the two the parity-free engine cannot serve. At $m = 2$ the engine is silent: the always-a-square inequality degenerates ($(m - 2)^2 = 0$) and the clearing interval $(1, 3]$ is genuinely square-free. At $m = 4$ the engine convicts every class from the outside: the failure window covers $r \in \{1, 2, 3\}$, and the $r = 0$ interval $(4, 8)$ is square-free. Outside view and inside view agree because both are reading the same two table rows: $\mathcal{Q}(2m) \subseteq \{0, 1, m\}$.

Theorem 6.3 (The confined classes are Golomb's game)**.** *Let $m \in \{2, 4\}$ and $0 \le r \le m - 1$, and let h be the least representative of the hard residue: $h = m + r$ if $r \ge 1$, and $h = 0$ if $r = 0$. Index the confined class of $D(m, r)$ by $n = 2mt + h$, $t \ge 0$, and let G denote the set of P-positions of Golomb's subtract-a-square game. Then for $m = 2$ the position $4t + h$ is P in $D(2, r)$ if and only if $t \in G$, and for $m = 4$ the position $8t + h$ is P in $D(4, r)$ if and only if $\lfloor t/2 \rfloor \in G$ — the confined class consisting of two interleaved copies of Golomb's game, one on each parity of t.*

Proof. (a) The three kinds of square move. Every $k \ge 1$ is odd, $\equiv 2 \pmod 4$, or a multiple of 4; we dispatch each kind from a hard-class position $n = 2mt + h$. If k is odd then $k^2 \equiv 1 \pmod 8$, hence $\equiv 1 \pmod{2m}$, and the landing $\equiv h - 1 \pmod{2m}$ is the odd rung directly above the foreign bottom $r - 1$ (bottom $m - 1$ when $r = 0$), an N-position by Lemma 3.1. If $k \equiv 2 \pmod 4$ then $k^2 \equiv 4 \pmod 8$: at $m = 2$ this is $\equiv 0 \pmod 4$ and the move stays in the class, while at $m = 4$ the landing $\equiv h - 4 \pmod 8$ is the free residue (r, or m when $r = 0$), every member of which is N because it subtracts 1^2 onto the foreign bottom directly beneath it — P by Lemma 3.1, legal from the smallest free member. If $4 \mid k$ then $2m \mid k^2$ and the move stays in the class. In the index t the internal moves read: at $m = 2$, every even $k = 2j$ subtracts $4j^2$, i.e. $t \mapsto t - j^2$; at $m = 4$, every $k = 4i$ subtracts $16i^2$, i.e. $t \mapsto t - 2i^2$. Legality transfers exactly: since the amounts and $2mt$ are multiples of $2m$ while $0 \le h < 2m$, we have $4j^2 \le 4t + h \iff j^2 \le t$ and $16i^2 \le 8t + h \iff 2i^2 \le t$. So the internal option sets at index t are $\{\, t - j^2 : j^2 \le t \,\}$ at $m = 2$ and $\{\, t - 2i^2 : 2i^2 \le t \,\}$ at $m = 4$ — verbatim the option sets of subtract-a-square and of subtract-$\{2i^2\}$. The three-way sort is genuinely three-way only at $m = 4$: modulo 4 the values $k^2 \equiv 4$ and $k^2 \equiv 0$ collapse into one residue, so at $m = 2$ parity alone decides — every even square stays internal — which is why $m = 2$ hosts Golomb's game whole while $m = 4$, whose middle class defects to the free residue, keeps only the doubled cousin subtract-$\{2i^2\}$. *(b) Terminals align.* At $m = 2$ the index game has the single internal-terminal $t = 0$; at $m = 4$ it has $t = 0$ and $t = 1$ (the smallest internal move needs $2 \le t$) — matching the terminal sets of subtract-a-square and subtract-$\{2i^2\}$ exactly. For $r \ge 1$ these positions have real moves, but by (a) every one of them exits onto an N-position; for $r = 0$ the alignment is literal: $h = 0$, so $t = 0$ is $n = 0$, a moveless position — Golomb's own terminal in the flesh. *(c) The transfer.* The status of a position is determined by its options alone, and by (a) every exiting option is N; so a hard-class position is P iff all of its internal options are N, and N iff some internal option is P — exits can neither supply a P-witness nor block one. By induction on t, the map $t \mapsto \text{status}(2mt + h)$ therefore satisfies exactly the recursion of the subtraction game with the option sets of (a). At $m = 2$ that recursion is Golomb's, proving the first claim. *(d) The $m = 4$ split.* In subtract-$\{2i^2\}$ every move is even, so the parity of t is invariant, and on each parity class the move $t \mapsto t - 2i^2$ reads $u \mapsto u - i^2$ in the half-index $u = \lfloor t/2 \rfloor$ (legality $2i^2 \le t \iff i^2 \le u$ on both parities), with terminal $u = 0$ on each side: two interleaved, independent copies of Golomb's game, so $8t + h$ is P iff $\lfloor t/2 \rfloor \in G$. $\square$

The anomaly ledger now reads as a codebook: $D(2,1)$'s recorded fingerprints $3, 11, 23$ are $4t + 3$ over $t = 0, 2, 5$ — the opening P-positions of Golomb's game (OEIS A030193) — and $D(4,1)$'s $5, 13, 37$ are $8t + 5$ over $t = 0, 1, 4$, half-indices $u = 0, 0, 2$: the same openers arriving through the two interleaved copies. The instrument had been filing Golomb's sequence under two aliases for the entire census — while re-deriving that very sequence as a calibration classic before every session.

Remark 6.4. Nothing here is special to squares: whenever a mode class of step $2m$ is confined, its internal game is the plain subtraction game on the index t with subtraction set $\{\, s/2m : s \in S,\ 2m \mid s \,\}$, where S is the mode's move set.

Corollary 6.5 (No eventual period)**.** *For $m \in \{2, 4\}$ and every r, the set of P-positions of $D(m, r)$ is not eventually periodic.*

Proof. Suppose some period p worked for all $n \ge N_1$. Membership would still repeat along any arithmetic progression, so restrict to the hard class $n = 2mt + h$: adding $\operatorname{lcm}(p, 2m)$ to n preserves both the class and, beyond N_1, the P/N status, and in the index this says the set $\{t : 2mt + h \text{ is P}\}$ is eventually periodic with period $q = \operatorname{lcm}(p, 2m)/2m$. At $m = 2$ that set is G itself, the P-set of Golomb's game, by Theorem 6.3. At $m = 4$ it is $\{t : \lfloor t/2 \rfloor \in G\}$, and one parity suffices: restrict to even $t = 2u$; membership of u in G repeats when t moves by $2q$, and $2q$ is a multiple of q, so G is eventually periodic with period q as well. Either way we are left holding: G is eventually periodic with period q beyond some N_0. Two facts kill this. The argument is folklore for any subtraction set containing a multiple of every integer; we include it to keep the paper self-contained. First, the P-positions of Golomb's game never run out — one losing position per horizon. Suppose no P-position existed at or beyond some M. Take $v \ge M/2$ and $t = (v + 1)^2 - 1$, which lies beyond M. Every legal square from t is at most v^2, because $(v + 1)^2$ overshoots t by exactly 1; and $v^2 \le t - M$, because $t - M = v^2 + 2v - M$

and $2v \geq M$. So every option of t lands at $t - k^2 \geq M$, all N by supposition, making t itself a P-position at or beyond M — contradicting the supposition. (The -1 in the construction does real work: at $t = (v+1)^2$ the square $(v+1)^2$ is legal, the move lands on 0, a P-position, and the position is honestly N.) Second, the kill. The first fact supplies a P-position t of Golomb's game with $t \geq N_0$. Periodicity walks membership up t's own progression: $t + q \in G$, then $t + 2q \in G$, and so on, every intermediate point lying at or above N_0, so every step is licensed. After qk^2 steps we reach $t + (qk)^2 \in G$. But $(qk)^2$ is two things at once: a multiple of q, which is what let periodicity carry us there, and a perfect square, which makes the single move from $t + (qk)^2$ down to t legal. That move joins one P-position to another, which the definition of P forbids. No period exists. □

Corollary 6.6 (Sparse but never scarce)**.** *For $m \in \{2, 4\}$ and every r, the set of P-positions inside the confined class contains no two elements differing by a perfect square. Consequently it has asymptotic density zero, while its counting function up to n stays at least $\sqrt{n}/2 - O(1)$.*

Proof. The difference-free property is closure read twice: two P-positions of the class are both in squares mode, so if they differed by a square, the larger would have a legal move onto the smaller — a move from P to P, which is forbidden. Density zero is then the Furstenberg–Sárközy theorem [15, 16], which says exactly that a set of integers avoiding square differences has density zero. We claim no novelty at this bridge: for Golomb's game itself the observation and the connection are recorded by Eppstein [18], and through Theorem 6.3 our statement and his are one statement in two coordinate systems. The floor is Golomb's counting trick [2, 17], run on G and carried back through the reindexing. Every u in $[0, T]$ is either in G or wins by one square move onto some $g \in G$ below it; a fixed g can serve at most $\sqrt{T}$ positions this way, one per square, plus itself. So $T + 1 \leq |G \cap [0, T]| \cdot (\sqrt{T} + 1)$, giving $|G \cap [0, T]| \geq \sqrt{T} - 1$. At $m = 2$ the confined positions up to n correspond to indices up to $T \approx n/4$, so at least $\sqrt{n}/2 - O(1)$ of them are P. At $m = 4$ there are two copies of G, one per parity of t, living on disjoint sets of positions, so their floors add: each copy runs at $T \approx n/16$ and contributes $\sqrt{n}/4 - O(1)$, together $\sqrt{n}/2 - O(1)$ again. That both moduli land on the same constant is the small arithmetic accident $2/\sqrt{16} = 1/\sqrt{4}$. □

Corollary 6.7 (One density for the whole family)**.** *For every $m \geq 2$ and every r, the P-set of $D(m, r)$ has asymptotic density $(m-1)/2m$. For $m \geq 3$, $m \neq 4$ the correction to the periodic law is a finite set; for $m \in \{2, 4\}$ it is the confined class — infinite, of density zero, and slowly vanishing: by Corollary 6.6 it still holds at least $\sqrt{n}/2 - O(1)$ members up to n.*

Proof. For $m \geq 3$, $m \neq 4$, Theorem 5.1 lists the P-set as exactly $m - 1$ residue classes modulo $2m$ together with finitely many named positions, and $m - 1$ classes out of $2m$ carry density $(m-1)/2m$. For $m \in \{2, 4\}$ the same $m - 1$ classes arrive from the chains alone — Lemma 3.1 never needed the healthy hypothesis — the free residue contributes nothing, all of its members being N, and the confined class contributes density zero by Corollary 6.6. Written out at $m = 4$, where the accounting is least obvious: by Theorem 6.3 the confined P-set is $(16G + h) \cup (16G + h + 8)$, the even and odd parities of t; an affine map $u \mapsto 16u + b$ scales a counting function by the constant 16, so each image inherits density zero from G, and a union of two density-zero sets is density zero — two, not infinitely many, is what makes that last step safe. □

7 Further solved games

Example 7.1 ($D(5,2)$ solved)**.** The P-positions of $D(5,2)$ are exactly the $n \equiv 0,1,3,4 \pmod{10}$.

Proof. Let $S = \{\, n \geq 0 : n \equiv 0,1,3,4 \pmod{10} \,\}$; we verify the three conditions of Lemma 2.2. *(i) Terminals.* A squares-mode position — $n \equiv 2 \pmod 5$ — always has the legal move 1^2, since the smallest such position is 2; a fallback position has its only move -5 exactly when $n \geq 5$. The terminal set is therefore the set of fallback positions below 5, namely $\{0,1,3,4\}$ — the four chain bottoms — and all of them lie in S. *(ii) Closure.* Every element of S is $\equiv 0,1,3,4 \pmod{10}$, hence $\equiv 0,1,3,4 \pmod 5$: no candidate lies in class 2, so a candidate's only move is the fallback -5, whose landings occupy the residues $5,6,8,9 \pmod{10}$ — all outside S. (Legality plays no role here: a move that does not exist cannot violate closure.) No move from S returns to S. *(iii) Escape.* The complement of S consists of the residues $\{5,6,8,9\} \pmod{10}$ — the odd rungs of the four chains — together with $\{2,7\} \pmod{10}$, which is class 2. The odd rungs escape by the fallback -5, landing on $\{0,1,3,4\} \subseteq S$, legal because the smallest member of each class is at least 5. Class 2 escapes by its two keystones: from $n \equiv 2 \pmod{10}$ subtract 1^2, landing on residue 1, legal from the smallest member 2; from $n \equiv 7 \pmod{10}$ subtract 2^2, landing on residue 3, legal from the smallest member 7. Every position outside S has a move into S — and the verification consumes only the two instances $1^2 \equiv 1$ and $2^2 \equiv 4 \pmod{10}$; no general fact about squares is invoked. By Lemma 2.2, S is exactly the set of P-positions of $D(5,2)$: the law holds for every $n \geq 0$, with no preperiod — and the P-set coincides with the window of Definition 4.1 for the reason recorded in Remark 7.3. $\square$

Remark 7.2 (Verification against the workbook)*.* Prediction first, then the rows: residue-2 positions $(2,12,22)$ should be rescued by any legal square $\equiv 1$ or $9 \pmod{10}$, residue-7 positions $(7,17)$ by any square $\equiv 4$ or $6 \pmod{10}$, and every other square should be wasted.

- $n = 2$: only move $2-1=1$ (P). Predicted rescue 1^2 present ✓, nothing else in the row to waste.
- $n = 7$: $7-1=6$ (N), $7-4=3$ (P). Rescue 2^2 present ✓; the 1^2 is wasted exactly as the congruence says ($1 \notin \{4,6\}$ mod 10, landing $\equiv 6$).
- $n = 12$: $12-1=11$ (P), $12-4=8$ (N), $12-9=3$ (P). Primary rescue 1^2 ✓, and the spare $9=3^2$ makes its first appearance in the first row big enough to afford it.
- $n = 17$: $17-1=16$ (N), $17-4=13$ (P), $17-9=8$ (N), $17-16=1$ (P). Primary 2^2 ✓ and spare 4^2 ✓.
- $n = 22$: $22-1=21$ (P), $22-4=18$ (N), $22-9=13$ (P), $22-16=6$ (N). Primary 1^2 ✓, spare 3^2 ✓.

The check came back stronger than required: not only does the predicted rescue sit in every row, every wasted square is wasted for the predicted reason — its residue misses the survivor set, and every wasted landing is $\equiv 6$ or $8 \pmod{10}$, both N-residues. The congruence machinery reproduces the workbook letter-for-letter, and the 200,000 mechanical sweep stands as the bulk consistency evidence behind it.

Remark 7.3 (The $D(5,2)$ coincidence, resolved)*.* For any escaped game the P-set equals its window by construction: P-set = (chain P's) $\cup$ (class-r P's) $= T \cup \emptyset = T$. The equality fails exactly where escape does — the three exceptional games have P-set $= T \cup \{m+r\}$, and the diseased games' P-sets properly contain their windows.

Theorem 7.4 (Foursquare)**.** *Consider the game on $n \geq 0$: if $n \equiv 1 \pmod 4$, subtract any positive square $\leq n$; otherwise subtract $s \in \{3, 8, 9\}$, $s \leq n$; normal play. Then n is a* P*-position if and only if $n \equiv 0, 2, 4,$ or $6 \pmod{16}$, for all $n \geq 0$ with no preperiod.*

Proof of Theorem 7.4. Let $S = \{\, n \geq 0 : n \equiv 0, 2, 4, 6 \pmod{16} \,\}$; since S and its complement are unions of residue classes modulo 16, every position lies in exactly one of the two. We verify the three conditions of Lemma 2.2 for S. *(i) Terminals.* The positions 0 and 2 are standard-mode (both $\not\equiv 1 \bmod 4$) and every allowed move $3, 8, 9$ exceeds them, so both are terminal. Conversely, every standard-mode $n \geq 3$ has the legal move 3, and every square-mode n has the legal move 1^2, since square mode requires $n \equiv 1 \pmod 4$ and hence $n \geq 1$. The terminal set is therefore exactly $\{0, 2\}$, and both members lie in S. *(ii) Closure.* Every element of S is even, hence $\not\equiv 1 \pmod 4$, hence standard-mode, so its moves are among $3, 8, 9$. The moves 3 and 9 are odd, and even minus odd is odd, so their landings occupy odd residues modulo 16 — all outside the all-even S. The move 8 carries the residues $\{0, 2, 4, 6\}$ onto $\{8, 10, 12, 14\}$, also outside S. (Legality plays no role here: a move that does not exist cannot violate closure.) Hence no move from S lands in S. *(iii) Escape.* The complement of S consists of the even residues $\{8, 10, 12, 14\}$ together with the odd residues, and the odd residues split by mode: $\{3, 7, 11, 15\}$ ($\equiv 3 \bmod 4$) are standard, $\{1, 5, 9, 13\}$ ($\equiv 1 \bmod 4$) are square. The even non-candidates escape by subtracting 8, landing on $\{0, 2, 4, 6\} \subseteq S$, legal because the smallest member of each class is at least 8. The standard-mode odds escape by $3 \to 0$ (residue 3), $3 \to 4$ (residue 7), $9 \to 2$ (residue 11), and $9 \to 6$ (residue 15), legal at the smallest members $3, 7, 11, 15$ respectively and a fortiori above. The square-mode odds escape by the squares 1^2 and 3^2: subtracting 1 carries the residues $\{1, 5\}$ to $\{0, 4\}$, and subtracting 9 carries $\{9, 13\}$ to $\{0, 4\}$, legal since $1 \leq n$ whenever square mode applies and $9 \leq n$ on classes whose smallest members are 9 and 13. Every position outside S therefore has a move into S — and the verification consumes only the two instances $1^2 \equiv 1$ and $3^2 \equiv 9 \pmod{16}$; no general fact about squares is invoked. By Lemma 2.2, S is exactly the set of P-positions of Foursquare: the law $n \equiv 0, 2, 4, 6 \pmod{16}$ holds for every $n \geq 0$, with no preperiod. $\square$

Remark 7.5. Every odd square is congruent to 1 or 9 modulo 16: writing $(2m+1)^2 = 4m(m+1) + 1$, the product $m(m+1)$ is even, so $4m(m+1) \equiv 0$ or $8 \pmod{16}$. The proof above needs only the two instances $1^2 \equiv 1$ and $3^2 \equiv 9$; the general statement is recorded because the mechanism remark below leans on it. The full Grundy-value sequence of Foursquare is OEIS A396951.

The question of which moves may be added to a subtraction game without changing it is an old one: Austin answered it in 1976 for purely periodic subtraction games — for each move s already present, the move $p - s$ may be adjoined, since the reply s reverses it out and the period p absorbs the pair — and Ho, in 2015, gave the idea its modern name, expansion sets [13, 14]. Proposition 7.6 and Remark 7.10 answer the same question for a mode-switching game, at the level of P-positions: the additions Foursquare tolerates are exactly the odd moves and the governor's clones $\equiv 8 \pmod{16}$, and the parity-and-residue argument that certifies them does for mode-switching games the job that reversibility did classically.

Proposition 7.6 (Kadam's Extension)**.** *Let F' be Foursquare with its standard-mode move set enlarged from $\{3, 8, 9\}$ to every $s \equiv 3, 8,$ or $9 \pmod{16}$ with $s \leq n$, square mode unchanged. Then the P-positions of F' are exactly $n \equiv 0, 2, 4, 6 \pmod{16}$ — identical to Foursquare's. (Conjectured by Balaji Kadam, personal communication, July 2026.)*

Proof. Adding moves can only destroy terminality, never create it, so the terminals of F' lie among Foursquare's terminals $\{0, 2\}$, and both survive because the smallest member of the extended set is still $3 > 2$ — the terminal set is $\{0, 2\}$, inside the candidate $\{0, 2, 4, 6\} \pmod{16}$. From a candidate position — even, hence standard mode — every legal move of F' subtracts some $s \equiv 3, 8$, or $9 \pmod{16}$, and the closure argument of Theorem 7.4 consumed exactly two facts about s, its parity and its residue modulo 16, never its actual size, so it transfers verbatim to every member of the extended family: the $s \equiv 3$ and $s \equiv 9$ moves are odd and land on odd residues, none of which lie in the all-even candidate, while the $s \equiv 8$ moves land on $\{8, 10, 12, 14\}$ (mod 16), also outside — no move from the candidate returns to it. In the other direction, the extended move set contains the original one and square mode is untouched, so every escape witness of the original proof — the standard moves and the squares 1 and 9 — remains a legal move of F' with the same landing, and a move set that only grows cannot lose an escape: every position outside the candidate still has its move into it. The candidate therefore meets all three conditions of Lemma 2.2 in F' — terminals inside, closed against return, reachable from everywhere outside — so it is the P-set of F', and it coincides with Foursquare's. □

Proposition 7.7 (Which enlargements are harmless)**.** *Let a mode-switching game obey a law with no preperiod and period p, and enlarge the move set of one mode by an amount a. The law survives if and only if no P-position has another P-position exactly a below it, with both positions in the enlarged mode.*

Proof. Adding moves can only affect closure, never escape: every old witness remains legal with the same landing, so condition (iii) of Lemma 2.2 survives untouched, and condition (i) can only shrink the terminal set, which the hypothesis of no preperiod already places inside the candidate. So the law survives precisely when closure does. Forward: if no P-position of the enlarged mode sits a above another P-position, then no new option leaves a P for a P, closure holds, and Lemma 2.2 returns the same P-set. Converse: if some P-position n of that mode has $n - a$ also P, then n now has an option landing on a P-position, so n is N — the law fails, and it fails first at the smallest such n. □

Remark 7.8 (The scoping is not decorative). The criterion asks about P-positions of the enlarged mode, not about all P-positions. Enlarging $D(5, 2)$'s squares-mode set by the amount 2 leaves its law completely untouched — verified to 20,000 — even though 2 is a difference of two P-positions of the game ($2 = 2 - 0$ modulo 10), because class 2 contains no P-position from which the new move could ever fire. An unscoped criterion predicts a break here and is simply wrong.

Corollary 7.9 (Kadam's Extension, revisited)**.** *In Foursquare every P-position is standard-mode, so the scoping is invisible and the criterion reduces to a residue condition: an added amount is harmless exactly when it is not congruent modulo* 16 *to a difference of two elements of* $\{0, 2, 4, 6\}$*. Those differences are* $\{0, 2, 4, 6, 10, 12, 14\}$*, so the harmless amounts are exactly the odd ones and those* $\equiv 8 \pmod{16}$ *— the governor's clones. The enlargement of Proposition 7.6 to the full residue classes of* 3*,* 9 *and* 8 *lies inside this set, which is why the P-set cannot move. A sweep over every added amount from* 1 *to* 32 *matches the criterion's prediction in every case, and the breaks land where the converse says they must, at* $n = 2, 4, 6$ *and* 16 *for the amounts* $2, 4, 6$ *and* 16*. One outlier is instructive: adding* 10 *breaks first at* $n = 16$*, not* $n = 10$*, because* $10 = 16 - 6$ *needs the larger P-position to exist before the pair can fire — the converse promises the smallest violating position, not the smallest position congruent to anything.*

Remark 7.10 (The mechanism: why 16, and why $\{3, 8, 9\}$). Foursquare's board splits three ways, because square mode lives only at $n \equiv 1 \pmod 4$: the evens, the standard-mode odds $\equiv 3, 7, 11, 15 \pmod{16}$, and the square-mode odds $\equiv 1, 5, 9, 13$. Within the evens, sort the standard moves by parity and the game simplifies at a stroke: 3 and 9 land among the odds, never on the all-even P-set, so the single even move 8 governs the even world alone. That world under one jump of 8 is precisely the shape of the Chain Lemma (Lemma 3.1) — four chains by residue modulo 8, with bottoms $0, 2, 4, 6$ too poor to afford the jump, and a jump of size m minting a law of modulus $2m$ — so 8 mints 16, with P at the even rungs: $n \equiv 0, 2, 4, 6 \pmod{16}$, Theorem 7.4, its four residues revealed as chain bottoms; and running the law modulo 8 instead kills closure on the spot, $8 \to 0$ being a move from the candidate into itself. And 3 and 9 are not decoys — delete them and the law breaks at once, first at $n = 3$ — for on the standard-mode odds, where squares are illegal, they are the sole rescuers: residue 3's only witness is 3 (to bottom 0), residue 7's is 3 (to 4), residue 11's is 9 (to 2), residue 15's is 9 (to 6) — while the squares police only their legal home, the value 1 carrying residues $\{1, 5\}$ to bottoms 0 and 4 and the value 9 carrying $\{9, 13\}$ to the same two bottoms. Here is the number-theoretic heart at its true address: odd squares modulo 16 take exactly two values, 1 and 9 (Remark 7.5), and the second is load-bearing precisely at residues 9 and 13 — the square-mode residues, no others — while modulo 8 every odd square is 1 and those residues would be stranded: the same poverty of squares modulo powers of two that confines $D(2, \cdot)$ and $D(4, \cdot)$, one refinement up and just barely wealth enough. So "why $\{3, 8, 9\}$" has its complete answer — three moves, three jobs, zero waste: one governor $\equiv 8 \pmod{16}$ whose doubling is the modulus, plus a pair of odd rescuers whose residues cover all four standard-odd classes; and the number 9's double life upgrades rather than dissolves — rescuer in both modes, carrying $\{11, 15\}$ as a move and $\{9, 13\}$ as a square, the entire upper odd world leaning on the value 9.

The mechanism classifies tampering as well: an enlargement of the standard set leaves the P-set fixed exactly when every added move is either odd, landing, from any candidate position, on the all-N odd world, or $\equiv 8 \pmod{16}$, a clone of the governor — verified: 1, 5, and 24 preserve the law, 2, 4, and 16 break it at $n = 2, 4, 16$, and the maximal enlargement, every odd number plus the whole clone class, holds to 2,000. The extension of Proposition 7.6 adds precisely the residue classes of 3, 9, and 8 themselves, squarely inside the harmless family, so the conjecture is not merely true but forced — and busy rather than idle: below 3,000 the new moves serve as genuine witnesses at 1,490 positions, 746 of them odd, the smallest being $19 \to 0$, a new odd move rescuing its own residue. None of this replaces the verification proof, which remains the proof of record; it explains why the object that proof verified could not have looked any other way.

Proposition 7.11 (Odd-fallback collapse at $m = 2$)**.** (Part (i).) *For every odd integer $a \geq 1$, the set of* P*-positions (states n with $G(n) = 0$) consists precisely of all even integers. Consequently, the emergent period is* 2 *with a residue-set size of* 1.

(Part (ii).) *For all odd fallbacks $a, a' \geq 1$ and all states $n \geq 0$, the Grundy values satisfy $G_a(n) = G_{a'}(n)$.*

Proof. **Setup.** We fix $r = 1$: squares fire on odd n, and the fallback rule applies on even n, making the allowed move from n to $n - a$. Let $G(n)$ denote the Grundy value of state n, with $G(0) = 0$.

*Part (i) (*P*-positions = Even Numbers).* We proceed by induction on n.

Base case ($n = 0$). Terminal position, $G(0) = 0$, so 0 is a P-position.

Inductive step. Assume for all $k < n$ that $G(k) = 0 \iff k$ is even (equivalently, $G(k) > 0 \iff k$ is odd).

Case 1: n is even. The only available move is subtraction of the odd fallback a: $n \to n-a$. Since n is even and a is odd, the destination $n-a$ is strictly odd. By the induction hypothesis, $G(n-a) \neq 0$ (if $n-a \geq 0$), or the move set is empty (if $a > n$).

- If $a > n$: the move set is $\emptyset$, so $\text{mex}(\emptyset) = 0$ and $G(n) = 0$.
- If $a \leq n$: the set of reachable values is $\{G(n-a)\}$ with $G(n-a) \neq 0$, so $\text{mex}(\{G(n-a)\}) = 0$ and $G(n) = 0$.

In both cases $G(n) = 0$. Thus all even positions are P-positions.

Case 2: n is odd. The move set consists of square subtractions $n \to n - k^2$ for all integers $k \geq 1$ with $k^2 \leq n$. Choosing $k = 1$ gives a legal move to $n - 1$. Since n is odd, $n - 1$ is even, so $G(n-1) = 0$ by the strong induction hypothesis. Thus $0 \in \{G(n-k^2) : k^2 \leq n\}$, forcing the mex to be positive, so $G(n) \neq 0$.

Therefore $G(n) = 0 \iff n \equiv 0 \pmod 2$.

Part (ii) (Full Sequence Equivalence). **Even positions.** What is $\text{mex}(\{v\})$ when $v \neq 0$, and what does that force $G(n)$ to be for even n?

For even n, the reachable set is either empty (if $n < a$) or $\{G(n-a)\}$. By Part (i), since $n-a$ is odd, $G(n-a) = v > 0$. Now $\text{mex}(\emptyset) = 0$, and $\text{mex}(\{v\}) = 0$ whenever $v > 0$. Therefore $G(n) = 0$ for every even n, regardless of the specific value of v, and regardless of whether the move exists at all. The parameter a evaporates at every even step.

Odd positions. At odd n, split the available squares k^2 by parity: which earlier positions' Grundy values enter the mex?

For odd n: if k is even, k^2 is even, so $n - k^2$ is odd; if k is odd, k^2 is odd, so $n - k^2$ is even. By Part (i), for every odd k the landing $n - k^2$ is even with $G(n-k^2) = 0$. Thus the reachable Grundy values from odd n are $\{0\} \cup \{G(n-k^2) : k \text{ even}, k^2 \leq n\}$. The value 0 is guaranteed by the odd square moves ($k = 1, 3, 5, \dots$), and the non-zero entries depend exclusively on $G(n-k^2)$ for even k — landings on smaller odd numbers. So G at odd n is computed purely from earlier odd states, completely bypassing all even positions, and thus bypassing a entirely.

By induction on n: $G(0) = 0$; every even state yields $G = 0$ independent of a; and every odd state's value is computed from prior odd states via even-square subtractions, in which a nowhere appears. The parameter a never impacts the Grundy sequence, so all odd fallbacks collapse to the exact same game. $\square$

8 Discovery methodology and retrodictions

The conjectures in this paper were proposed by GameHunter, a closed loop of four parts: a proposer that varies rulesets, a Grundy engine that computes exact values, deterministic detectors that flag periodicity and modular structure, and a novelty filter that discards sequences already catalogued in the OEIS. Seeded evolutionary hunts surfaced the earliest solved games, Foursquare among them; systematic sweeps then enumerated the diagonal family, 1,120 games in the first sweep and 1,848 by the last, with exact laws read in 842 of them. Every law survived six adversarial verification passes at depths 20,000 and 10^6 without a failure, and a deeper rescan of the 731 resistant games converted only 6. Before and after every session the engine re-derives five classical laws — the subtraction game $\{1, 2, 3\}$ and Golomb's subtract-a-square among them — so that drift anywhere in the stack would announce itself in the classics first.

The division of labor is strict and the repository states it in writing: the instrument proposes and verifies mechanically and proves nothing, every proof here is human-written, and each numerical claim above regenerates from the seeded scripts at https://github.com/hamzagul07/gamehunter, where the integrity protocol and its full session logs also live. The protocol earns its keep rather than asserting itself: one drafting error — a subtly different game verified faithfully across three documents — was caught only because the discipline requires sentences to follow printouts, never precede them.

Twice the instrument recorded numbers that no theory then existed to explain, and both entries were later derived exactly: the unexplained P-positions $3, 11, 23$ in $D(2, 1)$ and $5, 13, 37$ in $D(4, 1)$ are the images of Golomb's opening losing positions under the reindexings of Theorem 6.3, and the recorded preperiod 14 for $D(9, 4)$ is one more than the rebel position 13 of Proposition 4.6. Both were readings before they were consequences, which is the strongest form of confirmation a proposing machine can offer: the prediction was on the record before the mechanism that implies it existed.

The closest antecedent we know is the work of Holshouser and Reiter on subtraction games whose move sets vary with the position [5], which anticipates the position-dependent format studied here while leaving the mode-switching family and its classification open.

9 What is closed, and what was always Golomb's

The question this section used to pose is answered inside the paper. The two columns the census could never fit — the confined classes of $D(2, r)$ and $D(4, r)$ — are Golomb's subtract-a-square game up to an affine change of coordinates (Theorem 6.3); they never settle into any eventual period (Corollary 6.5); and their P-positions thin toward density zero while never falling below the $\sqrt{n}/2$ floor (Corollary 6.6). The instrument's old verdict — lawless to 200,000, no modulus up to 200 — is upgraded from a reading to three theorems. The verdict was also, we can now say, slightly modest: the instrument's own density figures were carrying the answer all along. $D(2, 1)$ reads 0.2586 at 200,000 against the limiting $1/4$ of Corollary 6.7, and the excess, 0.0086, is precisely the confined class's contribution — alive, thinning on schedule, and held above its $1/(2\sqrt{n})$ speed limit by the floor.

What remains open no longer belongs to this paper; it belongs to Golomb's game, where it has lived since 1966. The true growth rate of Golomb's P-set is wedged between two proven bounds — the $\sqrt{n}$ floor (Corollary 6.6, second half) and the $o(n)$ ceiling that Furstenberg–Sárközy places on any square-difference-free set (Corollary 6.6, first half) — and our instrument reads a count near $n^{0.7}$ across its whole range: a reading, not a theorem, and one that must stay instrument-tier forever unless a genuinely new bound arrives, since no finite prefix can certify an exponent. The Grundy values of the same game (OEIS A014586) are more mysterious still [6, 7]. Problem A1 of the Games of No Chance list [8] asks how the structure of a ruleset constrains the structure of its periods; the diagonal family now hands that problem a complete worked answer on one side of the boundary and an honest pointer across it. One sentence about the far side, kept claim-free: the confinement mechanism is not special to squares — Remark 6.4 already shows the confined class plays the subtraction game $\{\, s/2m : s \in S,\ 2m \mid s \,\}$ — and its consequences, including which games can be made to appear inside mode-switching hosts, are developed in work in progress.

We still leave the reader $D(2, 1)$ as a parting gift. Subtract any square from an odd number; subtract 2 from an even one. A pencil reaches 3, 11, and 23 inside an afternoon — and this paper can now tell you what those numbers do forever: they never fall into a rhythm, and they grow rarer without ever running out. What it cannot tell you is the pace of the thinning. That question is older than this paper, and it is still open.

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